ModelIt!

Teacher's Guide

The Reef Is Bleaching

Why the World's Most Colorful Ecosystems Are Turning White

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-LS2-1, MS-LS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS2-1, MS-LS2-4

Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students build a model to analyze how changes in ocean conditions affect coral reef ecosystems.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Ecosystems are dynamic; disruptions to components can cascade through the system.
Crosscutting Concept	**Stability and Change** — Students identify conditions that maintain reef stability vs. changes that push the system toward collapse.

CAST Testing Alignment

MULTIPLE CHOICE:

Ocean temperature rises 2°C above normal for 6 weeks. Which is the most likely immediate effect on a coral reef?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain how rising ocean temperatures lead to a cascade of effects that can collapse an entire coral reef ecosystem.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Coral reefs cover less than 1% of the ocean floor but support approximately 25% of all marine species. Often called 'rainforests of the sea,' they are among the most biodiverse and economically valuable ecosystems on Earth.

The Coral-Algae Symbiosis

Coral polyps have a mutualistic relationship with microscopic algae called zooxanthellae. The algae live inside coral tissue, photosynthesizing and providing up to 90% of the coral's energy. In return, coral provides shelter and nutrients. The algae also give coral its vibrant colors.

The Bleaching Process

When water temperatures rise 1-2°C above the normal summer maximum for 4+ weeks, coral becomes stressed and expels its zooxanthellae. Without the algae, the coral's white skeleton shows through — hence 'bleaching.' Bleached coral isn't dead yet, but without its algae it starves and becomes vulnerable to disease.

Ocean Acidification

The ocean absorbs about 30% of atmospheric CO2. When CO2 dissolves in seawater, it forms carbonic acid, lowering pH. More acidic water makes it harder for coral to build its calcium carbonate skeleton. This is sometimes called 'osteoporosis of the sea.'

Global Reef Status

The Great Barrier Reef experienced mass bleaching events in 2016, 2017, 2020, 2022, and 2024. Scientists estimate that 50% of the world's coral reefs have already been lost since the 1950s. If current trends continue, 90% could be gone by 2050.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify ocean temperature, ocean acidity, zooxanthellae population, and coral health as system components.

E — Establish Relationships
Students determine that temperature and acidity both negatively affect zooxanthellae, which positively affects coral health.

V — Visualize & Model
Students build the coral reef model showing cascading effects from environmental stressors to ecosystem collapse.

E — Evaluate Outcomes
Students compare healthy reef, heat wave, and double threat scenarios.

R — Revise & Extend
Students add fish population or algae overgrowth to model trophic cascades.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This is the same reef — one year apart. What happened?"*

Watch For

Show before/after images. Let students react.

Slide 2: Learning Objectives

3 min

Say: *"Today we model why the most colorful ecosystems on Earth are turning white."*

Watch For

Pre-teach symbiosis with a quick partner example.

Slide 3: Big Question

3 min

Say: *"Coral reefs support 25% of all marine life. What happens if they're gone?"*

Watch For

Quick write: What do you already know about coral?

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to understand ecosystem collapse."*

Watch For

Preview the LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of a coral reef system?"*

Watch For

Emphasize the symbiosis between coral and zooxanthellae.

Slide 6: Activity 2: Connections

8 min

Say: *"Temperature doesn't directly kill coral — it disrupts the symbiosis first."*

Watch For

Draw attention to the INDIRECT pathway of effects.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what happens during a marine heat wave."*

Watch For

Students predict, then run. Compare single vs. double threat.

Slide 8: Discoveries

5 min

Say: *"Why is the combination of heat and acidity so devastating?"*

Watch For

Connect to real Great Barrier Reef data.

Slide 9: STEM Challenge

12 min

Say: *"You're marine scientists hired to save the Great Barrier Reef. What's your plan?"*

Watch For

Teams design science-based rescue plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Ocean Temperature, Ocean Acidity (pH)
- **INTERNAL:** Zooxanthellae Population, Coral Health

Relationships

- **Ocean Temperature to Zooxanthellae Population:** NEGATIVE (-) — When temperature rises above normal, coral expels its zooxanthellae, reducing their population.
- **Ocean Acidity to Coral Health:** NEGATIVE (-) — Higher acidity makes it harder for coral to build its calcium carbonate skeleton.
- **Zooxanthellae Population to Coral Health:** POSITIVE (+) — More zooxanthellae means more energy for coral through photosynthesis — up to 90% of its food.

Key Discoveries

- Coral depends on zooxanthellae algae for up to 90% of its energy
- When water gets too warm, coral expels its algae — that's bleaching
- Ocean acidification makes it harder for coral to build its skeleton
- The combination of heat AND acidity is far worse than either alone

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Coral is a plant or rock"	Coral is an ANIMAL — specifically, colonies of tiny polyps related to jellyfish. They build calcium carbonate skeletons but are very much alive.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bleaching means the coral is dead"	Bleached coral is stressed and starving but not necessarily dead. If conditions improve within weeks, zooxanthellae can recolonize and the coral can recover. But prolonged bleaching leads to death.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate change only affects land"	The ocean absorbs 90% of excess heat from global warming and 30% of CO2. Marine ecosystems are being dramatically affected by rising temperatures and acidification.